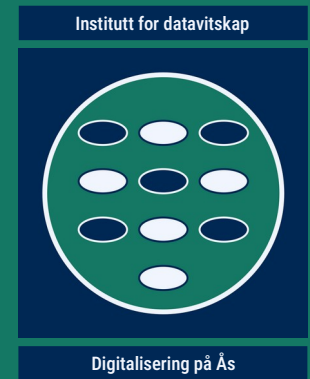
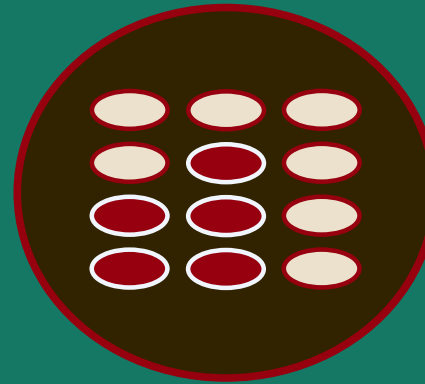


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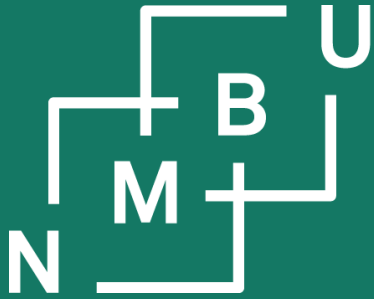


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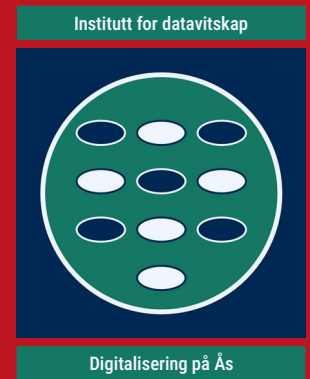
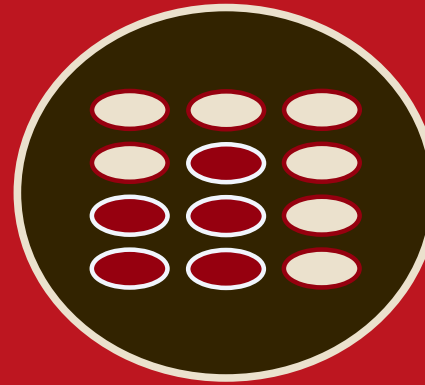
Datahandtering og -analyse

1 Ontologiar

- | | | | |
|-----|------------------------------|-----|-------------------------|
| 1.1 | Den semantiske veven | 1.4 | Ontologikutvikling |
| 1.2 | Vevontologispråket OWL | 1.5 | DOLCE, EMMO og BFO |
| 1.3 | Terminologikutvikling | 1.6 | Semantisk heterogenitet |



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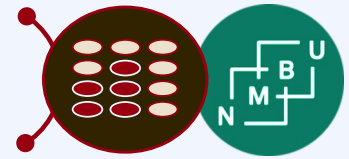


1 Ontologiar

1.3 Terminologi og taksonomi



Termliste



nn **multippel arv** *m.*

en **multiple inheritance**

nn **nærmaste overomgrep** *n.*

en **genus proximus**

nn **omgrepsskjema** *n.*

en **concept scheme**

nn **referent** *m.*

en **referent**

nn **taksonomi** *m.*

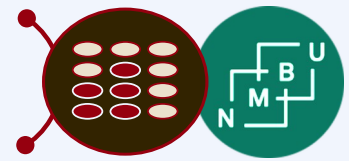
en **taxonomy**

nn **term** *m.*

en **term**

nn **åtskiljande kjenneteikn** *n.*

en **differentia**



Some advice from terminologists

There are real people who have terminology as their profession – I met some.

Their community does not engage with semantic technology so much – for example, they usually do not develop OWL ontologies. But they have good advice for us.

Distinguish the concept from the term.

The term is what we call the concept in some language.

It is good practice to have terms in multiple languages.

We must work more on this to keep all our languages fully functional.

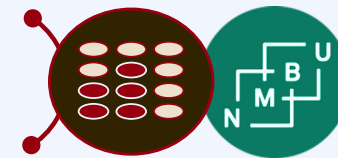
It is also OK to have multiple terms for the same concept in one language. Then it can be good to advise one form as the main form.

For us, the concept is the IRI and the owl:Class, the term is the label.

Terms should be in **lower case**, in **singular** number, and **not include an article**.

The definition should **not end with a full stop**. (It is not a sentence.)

Kva er det «terminologar» driv på med?



Definisjon – presisering av omgrepet

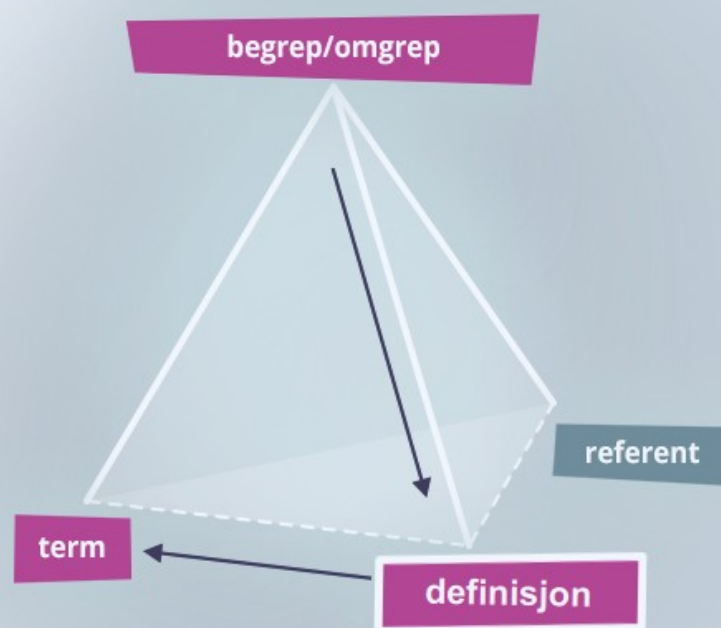
Ein definisjon er omgrepet formulert med ord. Definisjonen skal seie kva omgrepet tyder, og skilje omgrepet frå dei andre omgrepa som ligg rundt.

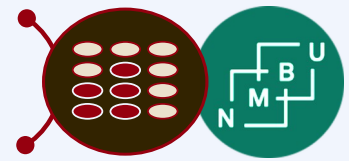
Ein definisjon i fagspråket er ofte meir presis enn ei forklaring i allmennspråket. Eit døme kan vere omgrepet 'planke'. I ei allmennordbok blir dette omgrepet gjerne forklart som 'langt stykke tre som er tjukkare enn eit bord og tynnare enn ein bjelke'.

I ei fagordbok vil ein kunne sjå ein definisjon som dette: 'trelast med tjukkeleik 38 millimeter eller større og breidde 75 millimeter eller større'. Definisjonen er meir presis enn forklaringa.



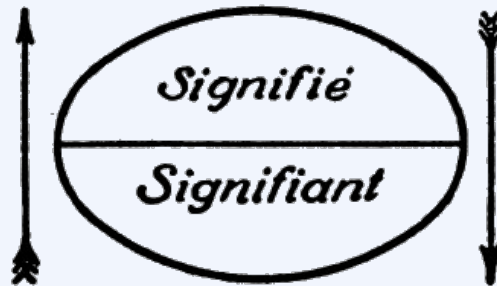
Pst, hugs at omgrep er det same som begrep på bokmål





Remark on the role of the referent

SAUSSURE

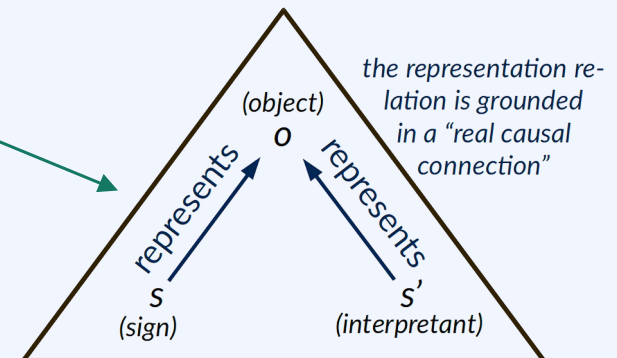


DERRIDA

«la lecture [...] **ne peut légitimement transgresser le texte** vers autre chose que lui, **vers un référent [...] ou vers un signifié hors texte** dont le contenu pourrait avoir lieu [...] hors de la langue, c'est à dire [...] hors de l'écriture
[...] quant à **l'absence du référent** ou du signifié transcendantal. *Il n'y a pas de hors-texte.*»

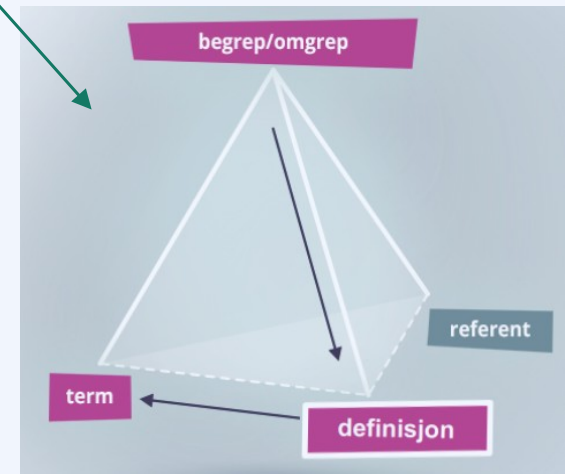
PEIRCE

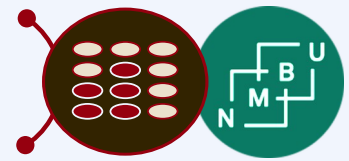
Peircean semiotics



the semiosis, a process by which a new representamen, the interpretant, is created

SPRÅKRÅDET





Engineering needs Aristotle

The core part of an **ontology** is the **terminology** and **taxonomy**.

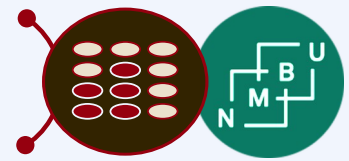
This is also where domain scientists are directly competent to contribute:

- Catalogue of concepts (what terms/concepts to include)
- Definitions of the concepts
- Taxonomy

A good way to define a concept is according to Aristotle, «An x is a y with the property z », where y is called genus ($\gamma\acute{\epsilon}\nu\omicron\varsigma$) and z differentia ($\delta\iota\alpha\phi\omicron\rho\acute{\alpha}$). These are called Aristotelian definitions or also genus-differentia definitions.

Aristotelian definitions accomplish to things:

- By writing up the definitions, we directly also create the taxonomy.
- Anything can directly be checked for what kind of thing it is, by testing, down from the most general category, what conditions it fulfills.



Aristoteliske definisjonar

Begrip begrepene!

Språkrådet

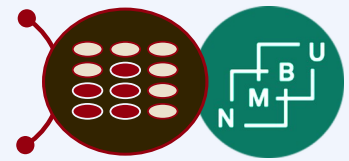
Begrip begrepene!

Formel på definisjon

Den vanlegaste måten å skrive definisjonar på er å ta utgangspunkt i generiske omgrepsrelasjonar. Difor bør du helst byrje med å teikne eit generisk omgrepssystem når du skal skrive definisjonar.

Når du har teikna eit generisk omgrepssystem, finst det ein formel for korleis du skriv ein god definisjon.

definisjon = nærmaste overomgrep + åtskiljande kjenneteikn



Retningslinjer for definisjonsskriving¹

- (6) ✓ mammal: **a vertebrate** that has hair, gives live birth, and nurses its young
- ✗ mammal: **a vertebrate** like, a cat, dog, or whale
- ✗ mammal: *Mammals* nurse their young.

3. Guideline: USE THE GENUS-DIFFERENTIA FORM

A definition should have the genus-differentia form, where the genus anchors the defined entity to a known entity at a higher level of generality, and the differentia (or differentiae) picks out those cases within this higher level

⁵ OBO Foundry Principles: Textual Definitions at <http://obofoundry.org/principles/fp-006-textual-definitions.html> [accessed: July 14, 2017].

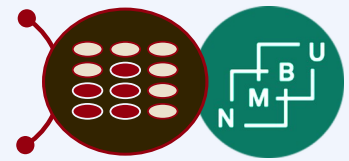
¹S. Seppälä, A. Ruttenberg, B. Smith, «Guidelines for writing definitions in ontologies», *Ciência da Informação* **46**(1): 73-88, doi:10.18225/ci.inf.v46i1.4015, **2017**

3.1. Guideline: INCLUDE EXACTLY ONE GENUS

A definition should always have one and only one genus (7). The genus is a superordinate term that tells us with what kind of thing we are dealing. Definitions of things like colors are expressed in the nominal form and start with a genus such as “a quality”. Similarly, definitions of processes and other entities that unfold in time, which are often expressed by verbs are, in ontologies, defined with nominal phrases.

3.1.1. Guideline: USE THE *GENUS PROXIMUS*

Use the *genus proximus*, that is, the closest parent term. This ensures that the genus is specific enough and that all the terms at the same level have the same genus.



Retningslinjer for definisjonsskriving¹

11. Guideline: Match Textual and Logical Definitions

To ensure consistency in ontology development and use, and to promote cohesion across definitions, it is recommended that the parts of a given textual definition match the parts of the corresponding logical definition as in (37).

- (37) ✓ bacteremia: An infection that has as part bacteria located in the blood. (IDO_0000506)
- bacteremia
EquivalentTo
infection and
(has_part some
(infectious agent and Bacteria and
(located_in some blood)))
-

¹S. Seppälä, A. Ruttenberg, B. Smith, «Guidelines for writing definitions in ontologies», *Ciência da Informação* 46(1): 73-88, doi:10.18225/ci.inf.v46i1.4015, 2017

Start by differentiating from the genus proximus

Knowledge graphs

Knowledge graphs contain individuals as nodes, contain relations as edges, and are understood as part of one semantic web.

Let us write this up as an Aristotelian definition with *genus* and *differentia*.

Start by differentiating from the genus proximus

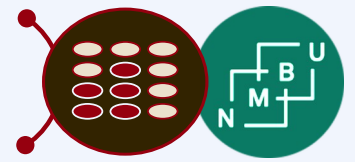
Knowledge graphs

Knowledge graphs contain individuals as nodes, contain relations as edges, and are understood as part of one semantic web.

Knowledge graph

A graph where the edge labels represent relations and the node labels are unique and represent individuals.

We can make the *genus* more specific based on what the *differentia* already presupposes. For example, instead of "lifeform with six legs," we might write "animal with six legs". Here, instead of "graph," we can write "labelled graph."



Minor style recommendations

Knowledge graphs

Knowledge graphs contain individuals as nodes, contain relations as edges, and are understood as part of one semantic web.

Knowledge graph

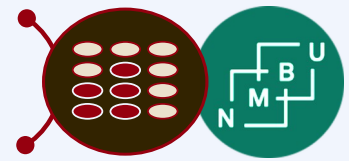
A graph where the edge labels represent relations and the node labels are unique and represent individuals.

knowledge graph

a labelled graph where the edge labels represent relations and the node labels are unique and represent individuals.

The term should be spelled in lower case.

The definition should also begin in lower case.



Minor style recommendations

Knowledge graphs

Knowledge graphs contain individuals as nodes, contain relations as edges, and are understood as part of one semantic web.

Knowledge graph

A graph where the edge labels represent relations and the node labels are unique and represent individuals.

knowledge graph

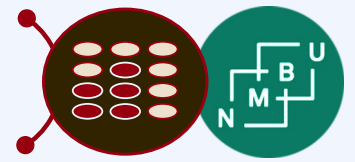
a labelled graph where the edge labels represent relations and the node labels are unique and represent individuals.

No article in the beginning!

No full stop in the end!

knowledge graph

labelled graph where the edge labels represent relations and the node labels are unique and represent individuals



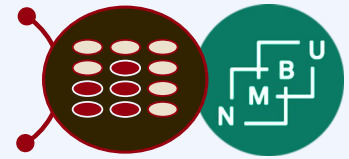
Discussion

Example from Wikipedia's glossary of graph theory:

cactus

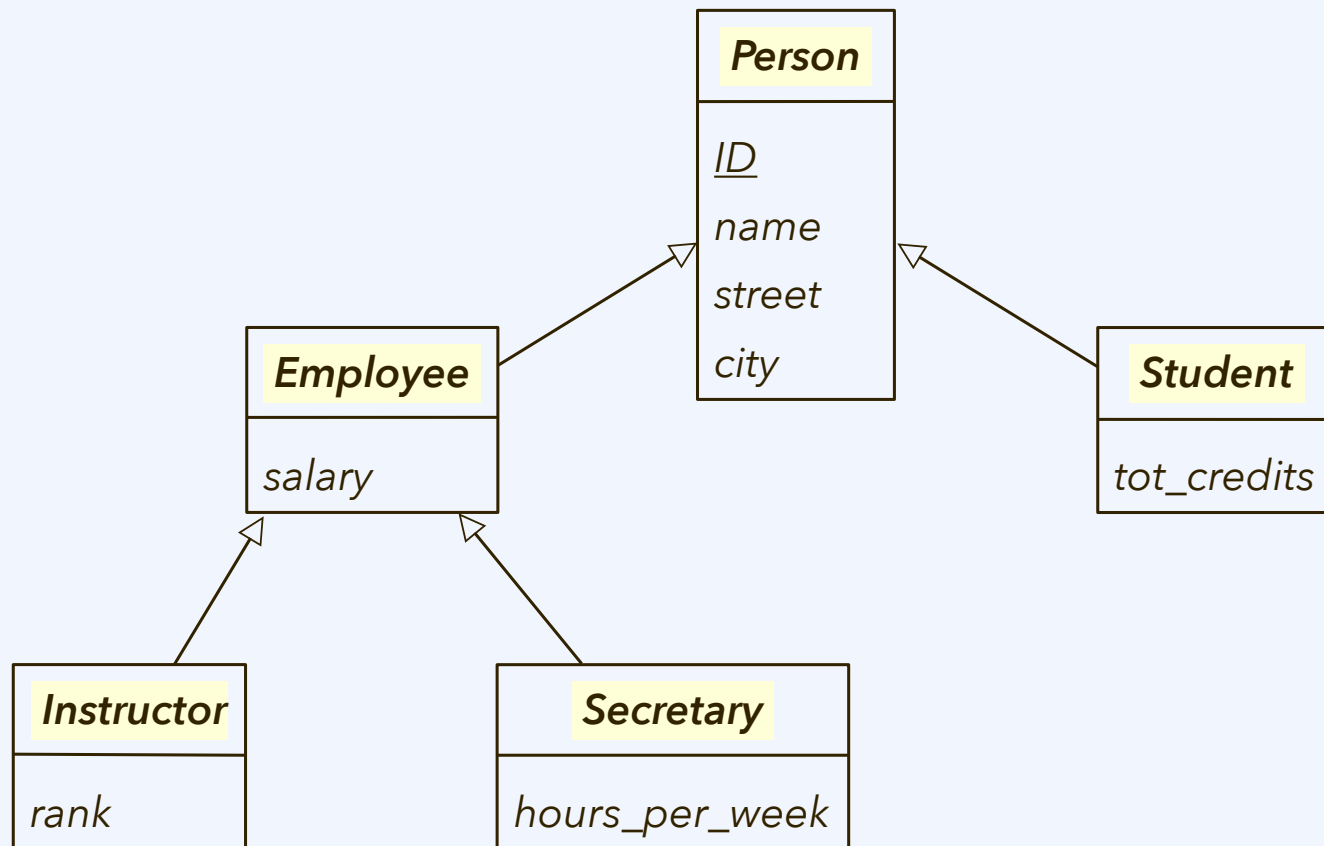
A cactus graph, cactus tree, cactus, or Husimi tree is a connected graph in which each edge belongs to at most one cycle. Its blocks are cycles or single edges. If, in addition, each vertex belongs to at most two blocks, then it is called a Christmas cactus.

What shall we do with this one?

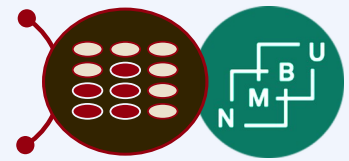


Taxonomy (conceptual hierarchy)

Example from Silberschatz *et al.*¹ (Fig. 6.18):



¹A. Silberschatz, H. F. Korth, S. Sudarshan, *Database System Concepts*, 7th int. stud. edn., McGraw-Hill, 2019.



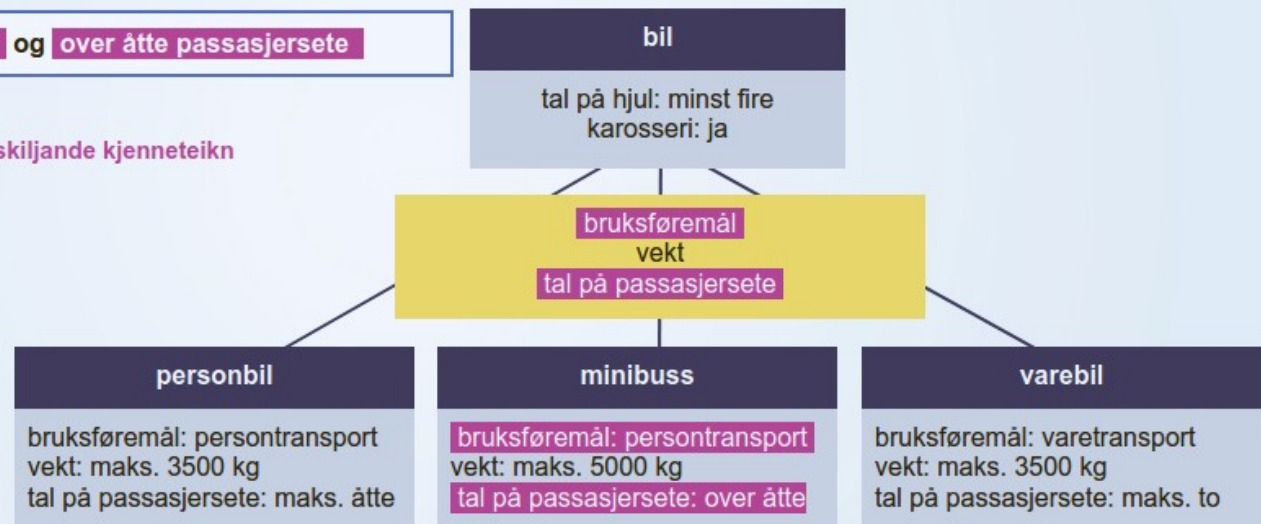
Taksonomi (omgrepshierarki)

Åtskiljande kjenneteikn 2/3

Ein annan type kjenneteikn vi fann, var *tal på passasjerseate*. Ein minibuss kan ha over åtte passasjerseate, medan ein personbil kan ha maks. åtte, og ein varebil berre kan ha maks. to.

minibuss = bil + persontransport og over åtte passasjerseate

definisjon = nærmaste overomgrep + åtskiljande kjenneteikn



Taxonomies and the genus-differentia method¹

20. Definitions should be unpackable.

Definitions should be substitutable for their defined terms without a change in meaning. If we define an A as “a B that Cs,” then we should be able to replace every occurrence of “an A” in a sentence with “a B that Cs,” and the result will have the same meaning (and thus also the same truth value) as the sentence with which we began. This process of substitution is called “unpacking.”

Thus in the FMA the reference of “heart” should be identical with the reference of the expression “organ with cavitated organ parts, which is continuous with the systemic and pulmonary arterial and venous trees.”

¹R. Arp, B. Smith, A.D. Spear, *Building Ontologies with Basic Formal Ontology*, MIT Press, 2015, Chapter 4: Principles of best practice II: Terms, definitions, and classification

Taxonomies and the genus-differentia method¹

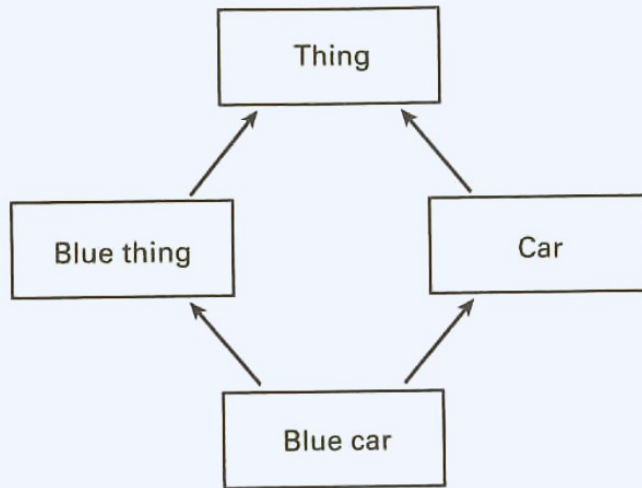


Figure 4.5

A simple illustration of multiple inheritance

Consider figure 4.5, which reflects the attempt to classify things using two different principles of classification (via color and via type of vehicle being classified). The figure does not satisfy the rule according to which (apart from the root) *every node within a hierarchy has exactly one parent*.

Here the principle of single inheritance is violated

¹R. Arp, B. Smith, A.D. Spear, *Building Ontologies with Basic Formal Ontology*, MIT Press, 2015, Chapter 4: Principles of best practice II: Terms, definitions, and classification

Alternatives to diamonds and ad-hoc unions

Imagine we have <Researcher> and <Engineer> in our ontology.

Without defining a diamond concept like ResearchEngineer, how can we say that Frank is a research engineer?

<Frank> a **<Researcher>, <Engineer>**.

If we additionally have <hasTeamMember> and <RegisteredTeam>, how can we say that every registered team must have at least one research engineer?

```
<RegisteredTeam> rdfs:subClassOf [  
    a owl:Restriction;  
    owl:onProperty <hasTeamMember>  
    owl:someValuesFrom  
        [ owl:intersectionOf (<Researcher> <Engineer>) ]  
].
```

Alternatives to diamonds and ad-hoc unions

Imagine we have <Researcher> and <Engineer> in our ontology.

We also still have <hasTeamMember>.

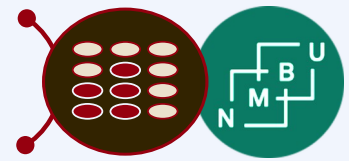
Without defining an ad-hoc union such as “ResearcherOrEngineer,” how do we say that only researchers or engineers can be team members?

```
<hasTeamMember> rdfs:range [  
    owl:unionOf (<Researcher> <Engineer>)  
    ].
```

How do we say: Frank is a researcher or an engineer, but I don't know which?

Answer: Don't say this. It is a bad idea to specify objects like this!!

But it *would* be <Frank> a **[owl:unionOf (<Researcher> <Engineer>)]**.



Behold – a man!

featherless biped

★ 2 languages ▼

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Etymology [\[edit\]](#)

WOTD – 12 February 2024

From the dialogue *The Statesman* by [Plato](#) (427–347 [B.C.E.](#)): “λέγω δὴ δεῖν τότε εὐθὺς τὸ πεζὸν τῷ δίποδι πρὸς τὸ τετράπουν γένος διανεῖμαι, κατιδόντα δὲ τάνθρωπον ἔτι μόνῳ τῷ πτηνῷ συνειληχὸς τὴν δίποδα ἀγέλην πάλιν τῷ ψιλῷ καὶ τῷ πτεροφυεῖ τέμνειν, [...] [I say, then, that we ought at that time to have divided walking animals immediately into biped and quadruped, then seeing that the human race falls into the same division with the feathered creatures and no others, we must again divide the biped class into featherless and feathered, [...]]”.^{[\[1\]](#)}



Diogenes - Behold A Man - A chicken as a featherless biped Throw Pillow

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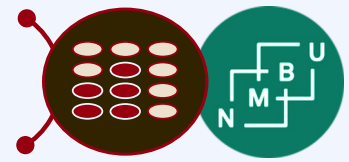
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Concept schemes using SKOS

particular:	object entity individual	relationship	property (sometimes: attribute)
universal:	class entity type concept	relation relationship type (in OWL: ObjectProperty)	attribute (sometimes: attribute type) (in OWL: DatatypeProperty)

Definition: A **concept** is a universal that is only instantiated by individuals.

- From **SKOS**, a semantic artefact for organizing concept schemes: “Concepts are the units of thought - ideas, meanings, or (categories of) objects and events - which underlie many knowledge organization systems” (Isaac & Summers 2009).
- In many settings, including in object-oriented programming, a concept is usually called a **class**. In E-R diagram terminology, it is called an **entity type**.
- E-R terminology distinguishes between an entity type and the corresponding **entity set**, i.e., the set of all individuals that instantiate the entity type. In nominalist ontology, these two are the same – a universal is the set of its individual instances.



Concept schemes using SKOS

SKOS for taxonomies

Look at the exact meanings of **skos:broader** and **skos:narrower**.

SKOS for alignments

Imagine you have a knowledge graph in ontology C (namespace prefix C).

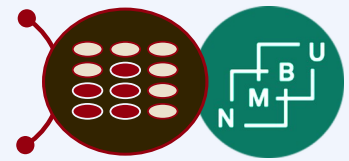
It is your task to transform it to ontology B (namespace prefix B). The target graph (knowledge graph obtained as solution) should only use B, no longer C.

The following are given in a SKOS alignment between B and C. Explain:

C:book **skos:exactMatch** **B:BookArtefact**.

C:author **skos:broadMatch** **B:hasCreator**.

C:person **skos:narrowMatch** **B:Author**.



Glossar

nn **nærmaste overomgrep** *n.*

snevraste omgrep frå eit omgrepsskjema (eller taksonomi, ontologi, osv.) som eit visst omgrep kan subsumerast under

- Det nærmaste overomgrepet blir brukt som fyrste ledd i ein omgrepsdefinisjon etter det aristoteliske genus-differentia-mønsteret.

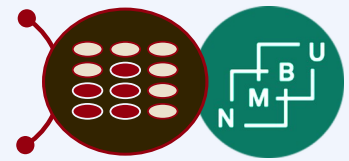
nn **omgrepsskjema** *n.*

system av universelle kategoriar (omgrep, forholdstypar o.l.) samt beskriving av koss dei heng saman (subsumsjon, liknande kategoriar, osv.)

nn **åtskiljande kjenneteikn** *n.*

kriterium som avgjer om eit visst individ som instansierer eit omgrep, også instansierer eit snevrare underomgrep

- Eit åtskiljande kjenneteikn blir brukt som andre ledd i aristoteliske genus-differentia-omgrepsdefinisjonar.



Glossar

nn **nærmaste overomgrep** *n.*; en **genus proximus**

most narrow concept from a concept scheme (or taxonomy, ontology, etc.) that a given concept can be subsumed under

- The *genus proximus* is used as the first element of a concept definition following the Aristotelian genus-differentia method.

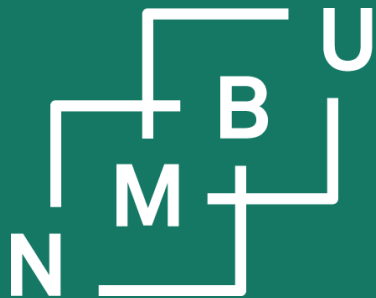
nn **omgrepsskjema** *n.*; en **concept scheme**

system of universal categories (concepts, relationship types, etc.) including a description of how they the universal categories are related to each other (such as by subsumption or similarity)

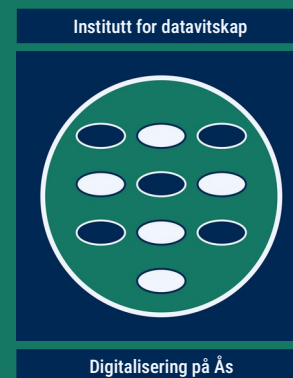
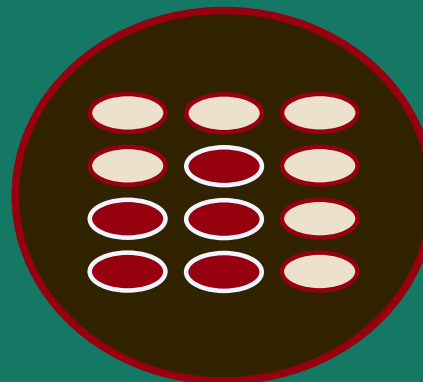
nn **åtskiljande kjenneteikn** *n.*; en **differentia**

criterion that determines whether a given individual, known to instantiate a certain concept, also instantiates a more narrow concept (i.e., a subclass)

- A definition that is formulated according to the genus-differentia pattern has the differentia as its second element, following the *genus proximus* (i.e., superclass).



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INF230

Datahandtering og -analyse

1 Ontologiar

1.1 Den semantiske veven

1.2 Vevontologispråket OWL

1.3 Terminologiutvikling

1.4 Ontologiutvikling

1.5 DOLCE, EMMO og BFO

1.6 Semantisk heterogenitet